U.S. ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

Test Operations Procedure 02-2-549 27 September 2023

MODELING AND SIMULATION (M&S) OF AUTONOMOUS UNMANNED GROUND VEHICLE (UGV) SYSTEMS

			Page
Paragraph	1.	SCOPE	2
	1.1	Purpose	2
	1.2	Applicability	3
	1.3	Limitations	3
	2.	FACILITIES AND INSTRUMENTATION	3
	2.1	Test Facilities and Condition	3
	2.2	Instrumentation	4
	3.	REQUIRED TEST CONDITIONS	5
	4.	TEST PROCEDURES	5
	4.1	Phase 1: Simulation Environment	6
	4.2	Phase 2a: Hardware-in-the-loop	8
	4.3	Phase 2b: Live Virtual Constructive (LVC) Simulation Test (Optional)	8
	4.4	Phase 3: Field Test	9
	4.5	Forced Failures	10
	5.	DATA REQUIRED	10
	6.	PRESENTATION OF DATA	11
APPENDIX	A	HARDWARE AND SOFTWARE SPECIFICATIONS	A-1
	B	SAMPLE SCORE SPACE	B-1
	C	GLOSSARY	C-1
	D	ABBREVIATIONS	D-1
	E	REFERENCES	E-1
	F	APPROVAL AUTHORITY	F-1

Approved for public release; distribution unlimited

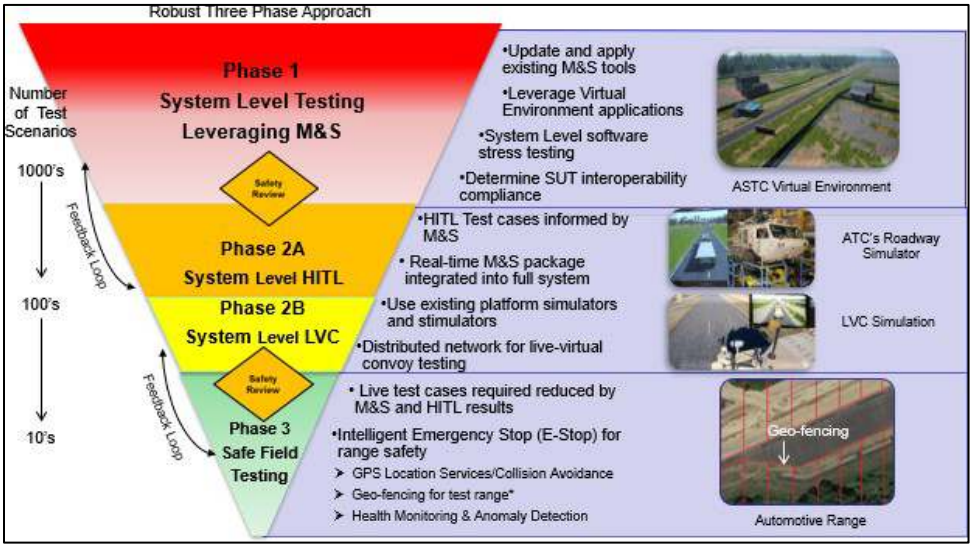
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

1. SCOPE.

This Test Operations Procedure (TOP) describes a three-phased modeling and simulation (M&S)-based approach to assess the safety and performance of autonomous unmanned ground vehicles (UGVs), referred to in this document as UGVs. UGVs are defined as a ground system platform with an autonomous controller software (ACS) stack that exercises authority over sensing, reasoning, and action execution without human intervention. The objective is to stress an autonomous system in a virtual and/or constructive environment to identify edge-case scenarios that lie within the desired performance and safety boundary to optimize the test and evaluation effort.

1.1 Purpose.

The purpose of this document is to present a hybrid M&S process to test UGVs across the full range of scenarios from completely virtual software testing, Hardware-in-the-loop testing, and elements of field testing. Figure 1 describes activities necessary to reduce the risks incurred when conducting field testing of UGVs, to include test selection that may have otherwise been overlooked in traditional design of experiment (DOE) planning, communications threat analysis, and more. This TOP details a methodology for developing and implementing M&S-based test scenarios to ensure the performance of the ACS stack meets stakeholder expectations. The intent is not to displace requirements-based testing, but to provide context. Observations resulting from this TOP can inform how close to failure the autonomy may be, while providing the scenario(s) that triggered key behaviors.



ASTC = Autonomous Systems Test Capability
HITL = Hardware-In-The-Loop
SUT = System Under Test
LVC = Live Virtual Constructive (test)

Figure 1. M&S Overview.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

1.2 Applicability.

a. The information contained herein applies to testing of autonomous and semi-autonomous ground systems. This document does not contain any guidance on licensure or recommendations for operating in the real-world environment.

b. This document supports testing of multiple UGVs working in concert to achieve mission success.

1.3 Limitations.

a. Methodologies for the testing of unmanned aerial vehicles or unmanned submersible vehicles are not covered in this document.

b. This document describes a program-agnostic process to testing UGVs. Mission-specific requirements must be considered and used to optimize the procedures laid forth in this TOP.

c. The results of testing from this document cannot be used on its own to validate UGV automotive performance. This TOP covers the testing of the decision making of autonomous SUT and should be used in conjunction with TOP 2-2-540 (app E, ref 1), Testing of Unmanned Ground Vehicles. The data resulting from these two TOPs should be considered holistically for evaluation.

d. Virtual testing to support evaluation is only valid under the conditions and limitations for which the M&S tools were verified, validated, and accredited (VV&A) in compliance with U.S. Army Test and Evaluation Command (ATEC) Regulation (AR) 73-21 (ref 2), Accreditation of Models and Simulations for Test and Evaluation, Section 8a., Documentation:

"Synchronization of test and evaluation (T&E) technology VV&A with overall T&E planning processes and documentation is critical, especially when T&E technology data is used in evaluation reports and safety assessments."

e. In addition to the limitations imposed by the VV&A of the M&S tool, it is the duty of the test officer to ensure diligent recordkeeping of all assumptions and data sources used to build the models, and to qualify result. Each input and output to/from the ACS stack needs to be documented and reported, as each are essential to test traceability.

2. FACILITIES AND INSTRUMENTATION.

2.1 Test Facilities and Conditions.

a. A software integration laboratory (SIL) should provide an enhanced lab-based virtual test environment, advanced M&S tools and reference libraries, local models and synthetic effects, and range connectivity to core test facilities to include dedicated high-performance computing clusters (app A).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

b. If a SIL is not accessible, HITL and LVC approach(es) should be considered before proceeding to field testing. At a minimum the virtual environment editor is required. This editor is the visual interface that enables scenario definition by way of actor placement, setting environment effects, and scripting.

c. To verify the UGV's decision making, vehicle test courses that are being used for UGV testing must be accurately digitally modeled when results will be used to compare against field test results. These course models may be digitally twinned using lidar and camera sensor fusion techniques and must be VV&A in accordance with AR 73-21. These facilities may be instrumented and calibrated to include meteorological monitoring stations to accurately depict changing conditions of the course.

d. Existing test facility types should be used when possible. The SIL instrumentation should be implemented where HITL operations occur. Operational equipment procedures should be created and will vary based on the type of simulation conducted at the facility. HITL facilities fill in the gap between software testing and field testing by offering safe, constrained vehicle-in-the-loop testing equipment. Examples include, but are not limited to, facilities equipped with a chassis dynamometer, roadway simulator, drivetrain simulators, etc.

e. Simulation state data (e.g., environment, vehicle, obstacles) should be recorded and used to ensure repeatability of the complete scenario as accurately as possible. This information can be stored in the form of a scenario configuration file.

f. The test events conducted in phases 2 and 3 should be recorded to document mishaps and incidents, while also providing situational awareness to personnel monitoring the test. Video coverage of the scenario should begin just prior to the beginning of the scenario and end when the scenario has concluded.

2.2 Instrumentation.

a. Instrumentation requirements for M&S testing of UGVs will rely on the capabilities and limitations imposed by the SUT's ACS stack. Automotive inputs that influence the decision-making of the ACS stack, yet not expressed virtually, must be documented in the Interface Control Document (ICD). The customer should explain the meaning of all inputs and outputs to the ACS stack and any special data reduction requirements. Determinate and indeterminate errors should be addressed. This will help the evaluation team determine the validity of the results.

b. Requirements are SUT-dependent and will be determined by the test officer in conjunction with the data required by the evaluator and/or customer. All vehicle bus data, to include but not limited to the Robotic Operating System and Controller Area Network, should be collected, as necessary.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

c. Additional instrumentation should be employed to measure the following, as applicable:

(1) All vehicles' 6D pose, which is its position and orientation on test course, acceptable error is one-fifth of system width. In addition, all acceleration and angular velocity data will be collected.

(2) Communication latency between internetworked systems (± 5 microseconds).

(3) Communication signal strength (decibels (voltage level) referenced to 1 microvolt per meter (dBμV/m)). The communications link bandwidth, modulation, signal quality, etc., should be collected, as available.

d. In general, instrumentation will continue to be used in traditional physical test roles to gather pertinent component-level data that are otherwise unable to be adequately simulated using accelerometers, thermocouples, spectrum analyzers, etc.

3. REQUIRED TEST CONDITIONS.

a. Prior to the start of testing, the test officer will run a checksum to verify that the virtual software version number being used matches the physical software version number.

(1) The ACS stack will be in a portable format with well-defined interfaces that is containerized or supplied in a virtual machine.

(2) Sensors that influence the ACS stack will be provided according to the manufacturer and model. Sensor models are developed from sensor specifications and are refined using real-world data.

(3) UGV platform model will account for vehicle characteristics from a low fidelity perspective including braking systems, suspensions, and powertrain systems. While high fidelity may be desired, it is not necessary to get an informed understanding of ACS stack decision-making. Again, models provided should undergo the VV&A process prior to use for evaluated tests.

b. The software and hardware versions for all subsystems will be recorded and remain unchanged during testing. If software or hardware revisions are made, the test officer may decide that test scenarios need to be repeated for the new configuration. Additionally, parameters that have been defined, either by a user or auto generated, will be recorded.

c. The SUT will be constrained to ensure safe operation while performing HITL testing.

4. TEST PROCEDURES.

a. Table 1 presents a list of parameters for simulated, hybrid, and field scenarios. This table should be used as a general guidance and scenarios should be added and/or removed, as necessary. Parameter definitions are presented in Appendix B. Test courses will be chosen based

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

on the scope of the test per the test officer and evaluator. The first scenario at each phase will include a straight maneuver on a dry road, in daylight, with no additional obstacles or forced failures. Further test scenarios will vary the weather, road conditions, light levels, and obstacles, as required for the mission environment. The decision to move into the next phase will be made at the agreement of the evaluator, program manager, and test officer. At that point, the test team should review the recommendations outlined in the test report to down-select critical scenarios.

TABLE 1. TEST PARAMETERS

Weather	Seasonal Effects	Time of Day	Road Type	Road Conditions	Obstacle	Forced Failure[b]
Clear	Soil Moisture	Dawn	Asphalt Pavement	Dry	Animal (S/D)	Sensor
Cloudy	Vegetation Density	Daylight	Dirt	Wet	Pedestrian (S/D)	Mechanical Messages
Fog	Vegetation Color	Dusk	Gravel	Snow	Vehicle (S/D)	Communi-cation
Rain (S/D)[a]		Night	Concrete Pavement	Ice	Signage	
Snow (S/D)		Twilight	Cobble-stone		Infrastructure	
Dust (S/D)					Vegetation	
Wind (Steady/Gust)					Negative space (e.g., ditches, potholes)	

[a] (S/D) denotes inclusion of both static (S) and dynamic (D) versions of specified obstacles.

[b] Parameters in the Forced Failure column are accompanied by additional instructions located in paragraph 4.4.

b. Based on the test goals and data required, test officers, subject matter experts (SMEs), and/or evaluators should decide which test parameters are needed. The number of runs required in the DOE is based on the test parameters selected and requested statistical power.

4.1 Phase 1: System-Level Testing Leveraging M&S.

a. Used to stress the ACS stack in a virtual environment beyond the safety and time constraints that typically limit field testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

b. The SUT will complete an initial simulation run in the SIL under clear conditions. No errors in the SIL fault log indicate that customer-provided models are compatible with the simulation environment. The intent of this initial simulation run is to verify compatibility between the simulation environment and the provided SUT model.

c. Assemble SUT's digital twin with appropriate mobility characteristics, model surfaces and materials, sensor models, and location(s). This model is a visual and mathematical representation of the UGV that can be engineered from a lidar scan and system specifications. Load the ACS stack onto a machine that meets the hardware requirements of that system.

(1) Use a stable version of the virtual environment to guarantee that any errors found can be directly associated with the SUT model.

(2) Create a scenario with a lightly overcast sky and no obstructions. The intent of this scenario is to ensure proper functionality before more exhaustive testing is conducted.

(3) Run scenario and ensure completion with no SIL-related faults. The decision to move into Edge Case Identification testing will be made at the discretion of the person responsible for the functionality and operations of the virtual environment, known as the test operator.

d. Edge Case Identification testing will be performed to identify critical scenarios where the ACS stack lie within the performance boundary within a threshold set by the test team. These results will be used to inform field testing.

(1) The test team (comprised of the test officer, test operator, evaluator, and program manager) will define the State Space. This is done by specifying the configuration parameters and ranges of values for the environment, the mission, and the SUT. The scoring parameters and constraints should also be identified.

(2) Scoring is based on a proximity to a performance boundary. A performance scoring matrix along with critical scenarios, also known as Score Space, will be identified by the test team. The team must identify the total number of scenarios for a given investigation; this is known as the Study Size. Once the study–that is the State Space, Score Space, and Study Size–is set, the test case can be executed. A sample score space visual is presented in Appendix B.

(3) Upon initiation by the test operator, the software application measures performance according to the Score Space and the virtual environment incrementally varies according to the State Space defined.

(4) Once the study is complete, identify scenarios with statistical significance to the test case. At the discretion of the test officer/test operator, the reported edge case scenarios will be replayed in the SIL for review.

(5) Repeat these steps and vary the scenario parameters, as needed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

4.2 Phase 2a: Hardware-in-the-Loop.

Used to assess how the ACS stack interacts with the UGV's physical platform in a controlled environment.

a. The test officer will identify which facility to use that will satisfy mission and data requirements. This facility should correspond to that which was selected for the virtual tests.

b. The test officer, test engineer, facility manager(s), and/or SME(s) will reference the SUT's ICD for integration with simulators.

c. Before introducing edge case scenarios identified in Phase 1, create a scenario to match ambient conditions as closely as possible.

d. The test operator will recreate the scenario in the virtual environment editor using the parameters of the edge case scenario found during virtual testing.

e. Disable, or override, the SUT's perception sensors and other external sensors, as required and feasible, on the physical platform for obstacle detection, and feed it a simulated signal that matches the virtual environment. Reroute radio communication to live SUT. This is done to stimulate the SUT by providing simulated inputs.

f. Test operator will operate the SUT's digital twin in the virtual world, while a field engineer or another test operator monitors the SUT on the facility platform.

g. Repeat with different edge case scenarios, as necessary.

4.3 Phase 2b: Live Virtual Constructive (LVC) Simulation Test (Optional).

Used to test a single UGV asset on available test track networked with virtual UGVs. At this phase, a safety operator is required. This may be in the form of a safety driver, who is in the vehicle, a person with the role of operating a safety-override mechanism, or technology. Forced failures of the Global Positioning System (GPS) and/or lidar at this stage are not advised without a proper risk assessment.

a. A system inspection will be conducted to verify basic automotive functions (acceleration, breaking, steering, etc.). During this inspection, verify the safe operation of the emergency stop (E-Stop).

b. The test officer will select the desired test facility according to mission and data requirements.

c. The test operator/field engineer will set up the test course and configuration files in accordance with the test parameters. Ensure communication links between live SUT and SUT's digital twin(s) is working properly prior to the start of the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

d. The safety operator will move the SUT into position. The test operator in the control station will confirm that the position and orientation is accurately reported to the SUT's digital twin.

e. After confirming positions, the safety operator in the live SUT will switch the vehicle controls into robotic operations mode. At this time, the SUT is operating in its autonomous state while the test operator at the control station maintains control of the digitally twinned SUT.

4.4 Phase 3: Field Test.

This phase requires a hardware applique kit that supervises the autonomous system during field testing, and cues soft and/or hard stops, if necessary. Before executing forced failures, risk assessment procedures must be conducted and completed. It is critical in this phase that field engineers tailor procedures to adhere to the appropriate TOP for the scenario or subtest being performed.

a. A system inspection will be conducted to verify basic automotive functions (acceleration, braking, steering, etc.). During this inspection, determine the functionality of the E-Stop.

b. The test officer will identify which facility will satisfy mission and data requirements.

c. The test operator will set the following metrics for the hardware applique kit at the control station:

(1) Operable geo-boundary.

(2) Allowable distance from objects.

(3) SUT channels to monitor, and limits on those to trigger a stop.

(4) Soft stop/hard stop triggers criteria. A soft stop is a request from the applique kit to stop an autonomous function. A hard stop is defined as an override full stop effort.

d. Mount the hardware applique kit to the autonomous system.

e. Set up a test course in accordance with the scenario parameters.

f. A test operator/field engineer must maintain visual contact with the SUTs, from either the chase vehicle or control station at all times.

g. Repeat with different scenarios, as necessary.

h. Subtest should be carried out in accordance with its applicable TOP. The instructions outlined in this section do not supersede procedures that must be carried out for each subtest.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

4.5 Forced Failures.

Failures of sensors, communications, and/or mechanical messages varying in severity cause many unforeseen impacts on autonomous system performance. The purpose of these test scenarios is to determine how the SUT will respond to various spontaneous failures.

a. Create an accurate virtual or field test scenario in accordance with the test specifications.

b. Simulate a failure in the test item appropriate to mission needs and assess its performance. Simulated failures include:

(1) Sensor failures, to include fuzzing and signal loss.

(2) Mechanical failures, to include flat tire, loss of cooling system, or suspension failure.

(3) Communication failures, to include packet loss, corrupted packets, and duplicate packets.

5. DATA REQUIRED.

a. Documentation will be provided detailing all aspects of testing, including, but not limited to, the following:

(1) Scenario identifier.

(2) SUT serial number.

(3) ACS stack version.

(4) SUT platform digital model.

(5) Photographs of test site and UGV model/system (as applicable).

(6) Any limitations of the vehicle or its subsystems during testing.

(7) Environmental conditions (light, visibility, temperature) and road types and conditions (paved, gravel, wet, sloped, etc.).

(8) Speed and position of SUTs and obstacles.

(9) Any failures of vehicle or its subsystems during testing.

(10) Predictive test case mission pass/fail report.

(11) Predictive test case safety pass/fail report.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

(12) SUT configuration.

(13) Virtual environment version and configuration.

(14) Version for perception system(s).

(15) Time (GPS clock).

(16) Obstacle (position, orientation, and scale) information.

(17) Network latency.

(18) Sensor fidelity.

b. When testing includes obstacles and forced failures, the following additional data should be collected and verified from the vehicle communication bus:

(1) Observation of obstacle/forced failure from vehicle communication bus.

(2) Observed response compared to required response.

c. Additional data should be collected based on unique features of the SUT. As an example: use the autonomy hardware applique kit (as detailed in the Phase 3: Field Test section), data identifying the observed corridor breach, and communications monitoring vehicle stops as triggered by the ACS stack.

6. PRESENTATION OF DATA.

a. Describe the inspection, specific test procedures, and results using narration, tables, photographs, charts, and graphs, as appropriate, or as outlined in the Test Procedures and Data Required sections.

b. Reduce, summarize, and analyze data from each subtest appropriate to the subtest data topic and failure definitions derived specifically for the item and the subtest category. Unique analytical tools (e.g., models, simulations, statistical techniques) should be described in sufficient detail to enable the reader to understand the basis for the analysis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

APPENDIX A. HARDWARE AND SOFTWARE SPECIFICATIONS

The proposed M&S approach has specific hardware and software requirements for creating and running simulations. This appendix covers those requirements. ATC has implemented this approach in their ASTC Laboratory.

a. Recommended Hardware.

(1) WS X299 SAGE ASUS Motherboard (with Trusted Platform Module (TPM) Header).

(2) TPM 2.0 Module.

(3) i9-10980XE Processor (24.75M Cache, 3.0-4.6 gigahertz (GHz), 18C/36T).

(4) 16-gigabyte (GB) DDR4-2666MHz Memory Modules.

(5) Intel 7600p Series 1-terabyte (TB) M.2 PCIe solid-state drive (SSD) (or Equivalent).

(6) EVGA RTX 2080 Super Graphics Cards, Blower Edition.

(7) E10G42BTDA 2-port 25GbE SFP+ LAN Adapter PCIe x8.

(8) MBD-H11SSL-NC Supermicro AMD EPYC Motherboard.

(9) EPYC 7551 Processor (64-MB Cache, 2.5-3.0 GHz, 32C/64T).

(10) 32-GB DDR4-2666MHz ECC/REG Memory Modules.

(11) Intel Pro 7600p SSDs, or equivalent.

(12) 12-TB Enterprise Capacity 3.5 hard disk drive.

(13) Intel Pro 7600p SSDs, or equivalent.

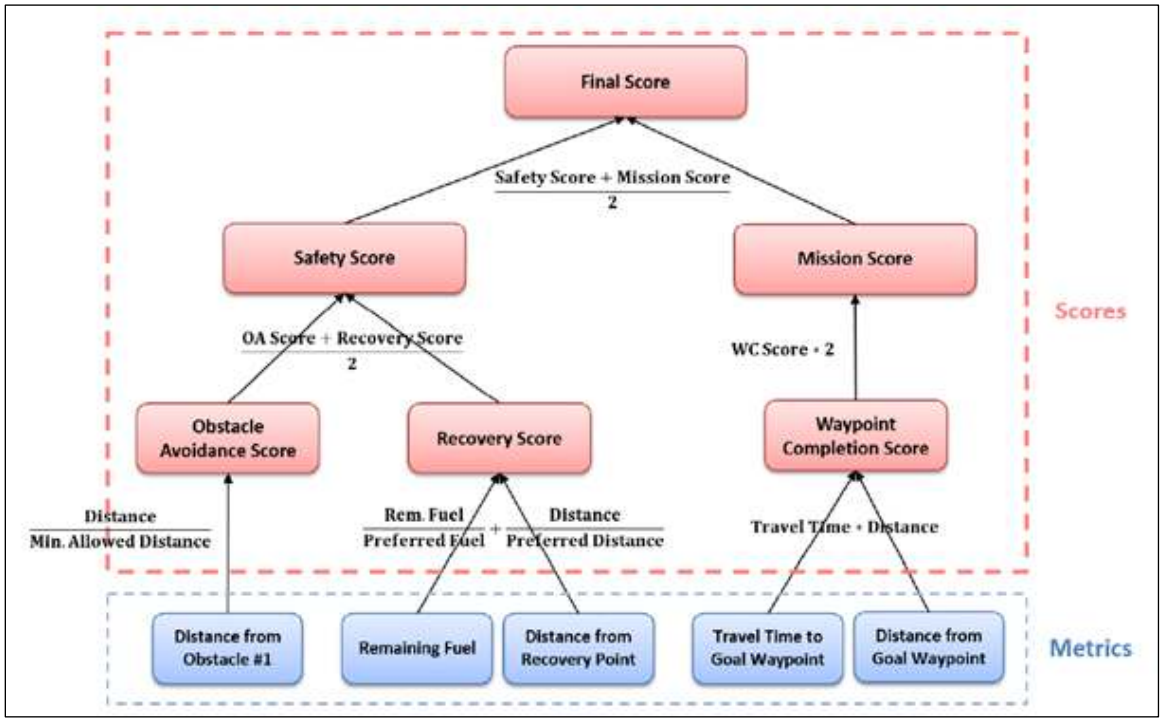
b. Recommended Software.

Unreal Engine	Open-Source
Range Adversarial Planning Tool	Johns Hopkins University Applied Physics Laboratory
Virtual Autonomous Navigation Environment	U.S. Army Corps of Engineers Engineer Research and Development Center

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

APPENDIX B. SAMPLE SCORE SPACE

State variables, or metrics, have been identified for a "capture the flag" simulation. A test operator has derived the following tree to define the simulation score space:



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

APPENDIX C. GLOSSARY

Term	Definition
Autonomous Controller Software	Software responsible for perception, navigation, and decision-making of a system.
Autonomous System	Physical platform equipped with autonomy controller software.
Autonomous UGV	A mode of control wherein the UGV is self-sufficient. The human operator can program a mission for the UGV, but the UGV will execute the mission without any human interaction.
Clear	No precipitation and no visibility obstructions.
Cloud	Visible aggregate of minute water droplets or ice particles above earth's atmosphere.
Dawn	The time of morning at which the sun is 6 degrees below the horizon. At this time, there is enough light for objects to be distinguishable and that outdoor activities can commence.
Daylight	Ambient light level of 10000 lux.
Dusk	The time at which the sun is 6 degrees below the horizon in the evening. At this time, objects are distinguishable but there is no longer enough light to perform any outdoor activities.
Dust	Loose dirt that is sent airborne by passing traffic.
Edge Case Scenarios	Critically ranked scenarios that lie along the mission and/or safety performance boundary.
Field Engineer	Working-level personnel conducting subtest execution.
Fog	Water droplets suspended in the air at the earth's surface. It is often hazardous when the visibility is reduced to one-quarter mile or less.
Forced Failure	Mission critical elements no longer working.
Gust	Rapid fluctuations in the wind speed with a variation of 10 knots or more between peaks and lulls. The speed of the gust will be the maximum instantaneous wind speed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

Hardware-in-the-loop (HITL)	Method of testing an SUT on specially equipped testbeds that simulate field conditions.
Heavy Rain	Liquid precipitation of more than 0.30 inches per hour (in./hr) Precipitation seems to fall in sheets. Individual drops are not identifiable. Spray of several inches is observed over hard surfaces.
Heavy Snow	Solid precipitation and reduced visibility to under one-quarter mile.
Live, Virtual, Constructive Simulation	Taxonomy to describe a scenario that has the combined involvement of real and virtual people and systems.
Light Rain	Liquid precipitation of 0.10 in./hr. From scattered drops that do not completely wet an exposed surface, to a condition where individual drops are easily seen.
Light Snow	Solid precipitation and reduced visibility no greater than one-half mile.
Moderate Rain	Liquid precipitation between 0.11 in./hr. to 0.30 in./hr. Individual drops are not clearly identifiable. Spray is observable just above pavements and other hard surfaces.
Moderate Snow	Solid precipitation and reduced visibility to between one-half mile and one-quarter mile.
Mostly Cloudy	When six-eighths to seven-eighths of the sky is covered by opaque (not transparent) clouds.
Night	Ambient light level less than 1 lux.
Partly Cloudy	Between three-eighths and five-eighths of the sky is covered by clouds.
Performance Boundary	An area in the Score Space where a discontinuity, or large change in performance, occurs.
Rain	Liquid precipitation of at least 0.10 in./hr.
Score Space	User-defined file that designates how scores are calculated from state variables.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

Semi-autonomous UGV	UGV that has multiple modes of control occurring simultaneously to include at least one autonomously controlled function. The level of semi-autonomy can vary greatly from UGV to UGV and will tend to be used extensively on weaponized UGVs (i.e., a weaponized UGV equipped with an Autonomous Navigation and Obstacle Avoidance System, but with tele-operated, operator-controlled weapon functions).
Snow (Weather)	Solid precipitation with reduced visibility.
Soil Moisture	The amount of moisture in the soil. It can be expressed in terms of volume or weight.
State Space	User-defined file that describes configuration parameters or metrics, and ranges of values for the environment, mission, and the vehicle, that are included in the evaluation of the SUT.
Steady	Sustained wind.
Twilight	Ambient light level of 10 lux.
Test Officer	The individual responsible for test conduct and data collection to pass to AEC evaluator.
Test Operator	See Field Engineer.
Vegetation Color	As the seasons change plants experience a variety of colors that have varying effects on sensor technologies.
Vegetation Density	The number of individual plants present per unit of ground area.
Wind	The horizontal motion of air.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

APPENDIX D. ABBREVIATIONS.

ACS	autonomous controller software
AR	ATEC Regulation
ASTC	Autonomous Systems Test Capability
ATC	U.S. Army Aberdeen Test Center
ATEC	U.S. Army Test and Evaluation Command
DOE	design of experiment
DTIC	Defense Technical Information Center
E-Stop	emergency stop
GB	gigabyte
GHz	gigahertz
GPS	Global Positioning System
HITL	Hardware-in-the-loop
ICD	Interface Control Document
M&S	modeling and simulation
S/D	static/dynamic
SIL	software integration laboratory
SME	subject matter expert
SSD	solid-state drive
SUT	system under test
T&E	test and evaluation
TB	terabyte
TOP	Test Operations Procedure
TPM	Trusted Platform Module
UGV	unmanned ground vehicle
VV&A	verified, validated, and accredited

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

APPENDIX E. REFERENCES.

1. TOP 2-2-540 Testing of Unmanned Ground Vehicles, 12 February 2009.

2. AR 73-21, Accreditation of Models and Simulations for Test and Evaluation, 10 July 2013.

For information only (related publications).

a. TOP 01-1-001 General Guidance for Incorporating Modeling and Simulation in Test Operation Procedures, 6 September 2004.

b. TOP 02-2-547, Creation of Autonomous Vehicle Testing Scenarios, 5 March 2020.

c. National Oceanic and Atmospheric Administration's National Weather Service Glossary. https://forecast.weather.gov/glossary.php.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

APPENDIX F. APPROVAL AUTHORITY

DEPARTMENT OF THE ARMY
U.S. ARMY WHITE SANDS MISSILE RANGE
100 HEADQUARTERS AVENUE
WHITE SANDS MISSILE RANGE, NEW MEXICO 88002-5000

TEWS-ED (73-1f) 27 September 2023

MEMORANDUM FOR SEE DISTRIBUTION

SUBJECT: Test Operations Procedure 02-2-549 Modeling and Simulation (M&S) of Autonomous Unmanned Ground Vehicle (UGV) Systems

1. Test Operations Procedure (TOP) 02-2-549 Modeling and Simulation (M&S) of Autonomous Unmanned Ground Vehicle (UGV) Systems has been reviewed by the U.S. Army Test and Evaluation Command (ATEC) Test Centers, the U.S. Army Operational Test Command, and the U.S. Army Evaluation Center. All comments received during the formal coordination period have been adjudicated by the preparing agency.

2. Scope of the document. This TOP describes a three-phased modeling and simulation-based approach to testing unmanned ground vehicles (UGVs), defined as a ground system platform with an autonomous controller software (ACS) stack that exercises authority over sensing, reasoning, and action execution for the system.

3. This document is approved for publication and will be posted to the Reference Library of the ATEC Vision Digital Library System (VDLS). The VDLS website can be accessed at https://vdls.atec.army.mil/otcs/cs.exe.

4. Comments, suggestions, or questions on this document should be addressed to U.S. Army White Sand Missile Range (TEWS-ED), 100 Headquarters Avenue - Suite 231 White Sands Missile Range, NM 88002-5000; or e-mailed to usarmy.apg.atec.mbx.atec-standards@army.mil.

LIDDIARD.VINCENT.MARK.1261726331
Digitally signed by LIDDIARD.VINCENT.MARK.1261726331
Date: 2023.09.27 17:55:30 -06'00'

VINCENT M. LIDDIARD
SES, USA
Executive Director

DISTRIBUTION
Commander, U.S. Army White Sands Missile Range
Executive Director, U.S. Army Evaluation Center
Commander, U.S. Army Operational Test Command
Commander, U.S. Army Yuma Proving Ground
Commander, U.S. Army Dugway Proving Ground
Commanders, U.S. ATEC Test Centers
Director, U.S. ATEC Tropic Regions Test Center
Director, U.S. ATEC West Desert Test Center

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

Forward comments, recommended changes, or any pertinent data which may be of use in improving this publication to the following address: U.S. Army White Sand Missile Range (TEWS-ED), 100 Headquarters Avenue - Suite 231, White Sands Missile Range, NM 88002-5000. Technical information may be obtained from the preparing activity: Virtual Test & Advanced Electronics Directorate (TEAT-VT), U.S. Army Aberdeen Test Center, 400 Colleran Road, Aberdeen Proving Ground, MD 2005-5059.

Additional copies can be requested through the following website: https://www.atec.army.mil/publications/documents.html, or through the Defense Technical Information Center, 8725 John J. Kingman Rd., STE 0944, Fort Belvoir, VA 22060-6218. This document is identified by the accession number (AD No.) printed on the first page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.